Anish Goel

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Professional Summary

Mechatronics Engineering undergraduate specializing in robotics, autonomous systems, embedded systems, and control engineering. Experienced in designing and building integrated robotic platforms across underwater, surface, aerial, and mobile domains. Proven ability to collaborate across mechanical, electronics, and software teams to deliver high-performance, system-level solutions. Seeking roles in robotics engineering and autonomous systems R&D.

Education

Thapar Institute of Engineering and Technology — **Aug 2023 – May 2027**
Bachelor of Engineering in Mechatronics (CGPA: 9.52) — *Patiala, Punjab*

Core Competencies

- Robotics Engineering
- Autonomous Systems
- ROS
- Embedded Systems
- Control Systems
- Autonomous Navigation
- Sensor Fusion
- Robot Control
- System Integration
- Mechanical Design
- Rapid Prototyping
- Robotics Research

Projects

Autonomous Underwater Vehicle (AUV) | *Patent No. 202611022081*
- Engineered a 1.5 m, 25 kg underactuated torpedo-shaped Autonomous Underwater Vehicle (AUV) featuring single-thruster propulsion and belt-driven fin actuation, optimizing hydrodynamic efficiency and low-drag maneuverability.
- Architected a modular structural assembly that supports a versatile 5kg payload capacity for diverse mission profiles.
- Contributed to embedded systems integration and fabrication, integrating onboard computing into a compact system.

Autonomous Surface Vehicle (ASV) | *In Development*
- Architected an Autonomous Surface Vehicle (ASV) for real-time environmental monitoring, utilizing a ROS-based navigation stack for precision river mapping.
- Designed 3D CAD model for a modular, waterproof chassis supporting diverse sensor payloads and aquatic stability.
- Programmed a ROS-based autonomy stack, leveraging sensor fusion and navigation nodes for end-to-end mission control and autonomous navigation.

AI-Integrated Tonometry System | *Patent No. 202511115771*
- Developed a low-cost intraocular pressure monitoring system to enhance early-stage glaucoma diagnostics.
- Architected the mechanical assembly and ergonomic housing, optimizing for optical alignment and portability.
- Integrated sensing and embedded electronics, reducing system cost by >80% compared to conventional devices.

Autonomous Aerial Robotics Platform | *Open-Source Prototype*
- Designed and modeled quadcopter chassis, performing thrust-to-weight and structural stability calculations.
- Optimized frame layout for payload integration and autonomous flight testing.

Self-Balancing Robot
- Engineered a self-balancing robotic system, leveraging 6-axis IMU sensor fusion for high-fidelity dynamic stability.
- Programmed a real-time PID controller with a 10 ms sampling rate (100 Hz), ensuring robust tilt stabilization.

Liquid Metal 3D Printing System | *Patent No. 202511034015, 202511099165*
- Led the design of an industrial-grade metal 3D printer, overseeing the integration of furnace-nozzle thermal systems and multi-axis CNC kinematics for direct-write deposition.
- Developed flow-mixing protocols for the additive manufacturing of multi-material and composition-graded structures.

Leadership Experience

Society of Intelligent Systems — **Feb 2025 – Present**
Project Lead — *Patiala, Punjab*
- Led a 25+ member multidisciplinary team (mechanical, electronics, software) to develop autonomous robotic systems.
- Drove system design reviews, aligning hardware, embedded and software modules to ensure seamless system integration.

Technical Skills

Software: C++, Python, C, Arduino IDE, ROS, MATLAB, Git, Maxsurf, Solidworks, Linux, AutoCAD, Eagle, Creo
Hardware: Arduino, ESP32, Raspberry Pi, GNSS/GPS Modules, Sensor Integration, Embedded Systems, PCB Prototyping, CAD & Mechanical Design, 3D Printing, Robotics System Integration

Awards

- Gold Medalist 2025 (Rank #1, Highest CGPA (2nd Year), Mechatronics)
- ROBO-AI Industrial Training (2024, 80 hrs): Robotics, Automation, AI (NVIDIA/MSFT/AWS)